

AUTOMATED BATTERY SWAP STATION

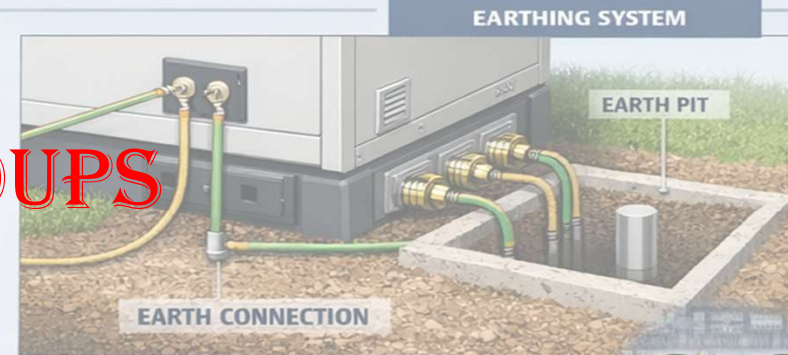


ELECTRICAL PANEL

AUTOMATED BATTERY SWAP

STATION

FROM S&S GROUPS



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Automated Battery Swap Station

Detailed Technical Description

① User Interface

10.1" Touch Display Panel

- Slot Availability
- Charging Status Monitoring
- Battery Swap Instructions
- User Authentication

① Connectivity

- 4G-IoT Based Connectivity
- RFID/NFC Access

② Battery Compartments

- 12 Lockable Battery Compartments
- Automatic Locking System
- LED Indicators for Charging Status
- Battery Availability
- Fault Conditions

① Safety Mechanisms

- Smoke Detection Sensor
- Water Level Detection Sensor
- Electrical Protection (Overtemperature Overcurrent)

Indian Standard Based

② Battery Compartments

- Automatic Locking System
- LED Indicators

Ⓐ User Interface

- 10.1" Display Panel
- User Authentication

② Connectivity Options

- 4G-IoT Connectivity
- RFID/NFC Access

⚠ Safety Mechanisms

- Smoke Detection Sensor
- Water Level Detection Sensor
- Electrical Protection (Overtemperature)
- Battery Availability, Fault Conditions

✓ Electrical Specifications

- Input Voltage: 150V – 270V AC
- Current Rating: 115 Amp
- Power Factor ≥ 0.92

⬇ Operating Features

- 0°C – 45°C
- Auto Cut-off Above 50°C

📏 Physical Dimensions :

- L: 1866 mm, W: 744, H: 1600 mm
- Weight: Approx. 360 kg
- Heavy-duty Powder Coated Steel Cabinet

764 mm

1866 mm

CRCA SHEET STEEL STRUCTURE
2mm BODY / 1.6mm DOORS

Automated Battery Swap Station

**INDUSTRIAL
LOCKING DOORS**

**LED STATUS
INDICATORS**

VENTILATION LOUVERS

**IP54 Rated
Weatherproof Enclosure**

**Heavy Duty
Sliding Battery Racks**

**Battery Slots
with Cooling Fans**

**Battery Slots
with Cooling Fans**

**Brass Double
Compression Cable Glands**

**BRASS DOUBLE COMPRESSION
CABLE GLANDS**

**Cable Gland Plates
at Bottom Entry**

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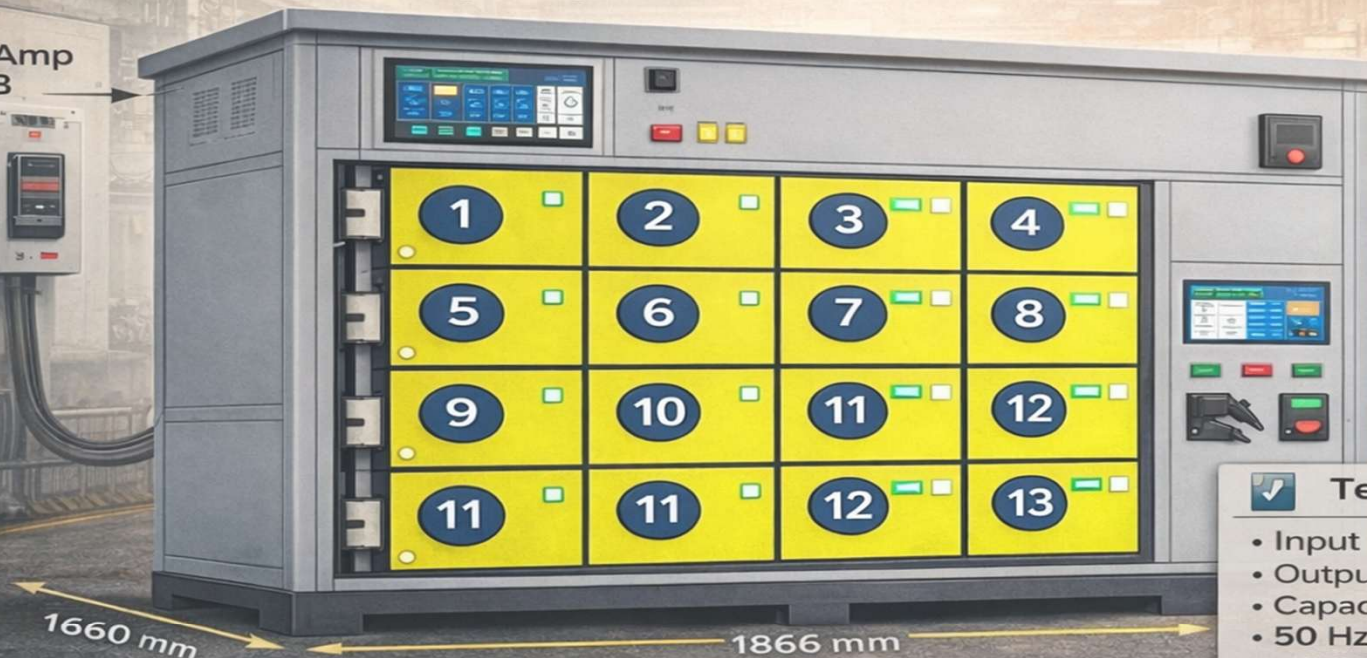
Battery Charging System

- Indian Standard Design
- 12 Battery Slots with Smart Charging
- Heavy-duty Powder Coated Enclosure



IoT Connected
Cloud Management

High Amp
MCCB



Technical Specifications

- Input 150V – 270V AC
- Output 230V DC / 15A each slot
- Capacity: 12 slots
- 50 Hz
- >0.92 Power Factor
- Approx: 360 Kg

Safety & Connectivity Specifications



Smoke & Water
Level Detectors



Overcurrent &
Overtemperature
Protection



RFID/NFC Card
Authentication

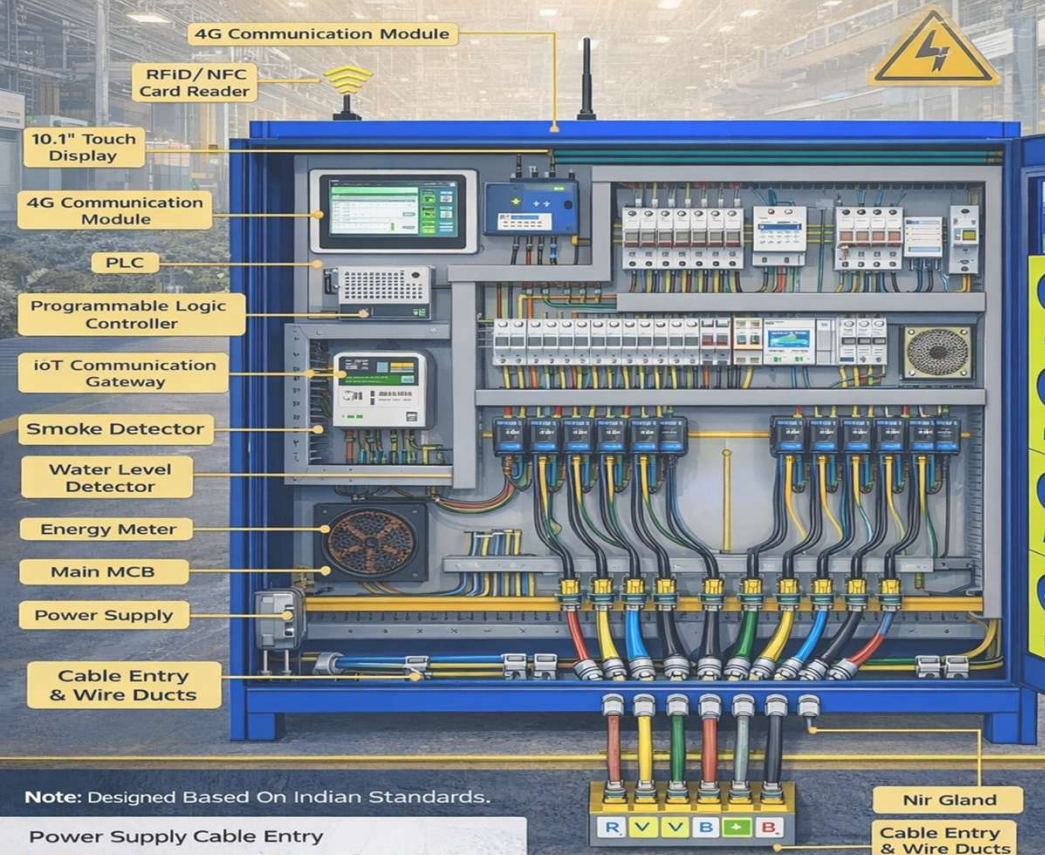


4G IoT & Cloud Connectivity
4G IoT & Cloud Connectivity

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AUTOMATION & CONTROL SYSTEM & INTERNAL WIRING & CABLE ENTRY VIEW

[INTERNAL ELECTRICAL SCHEMATIC]

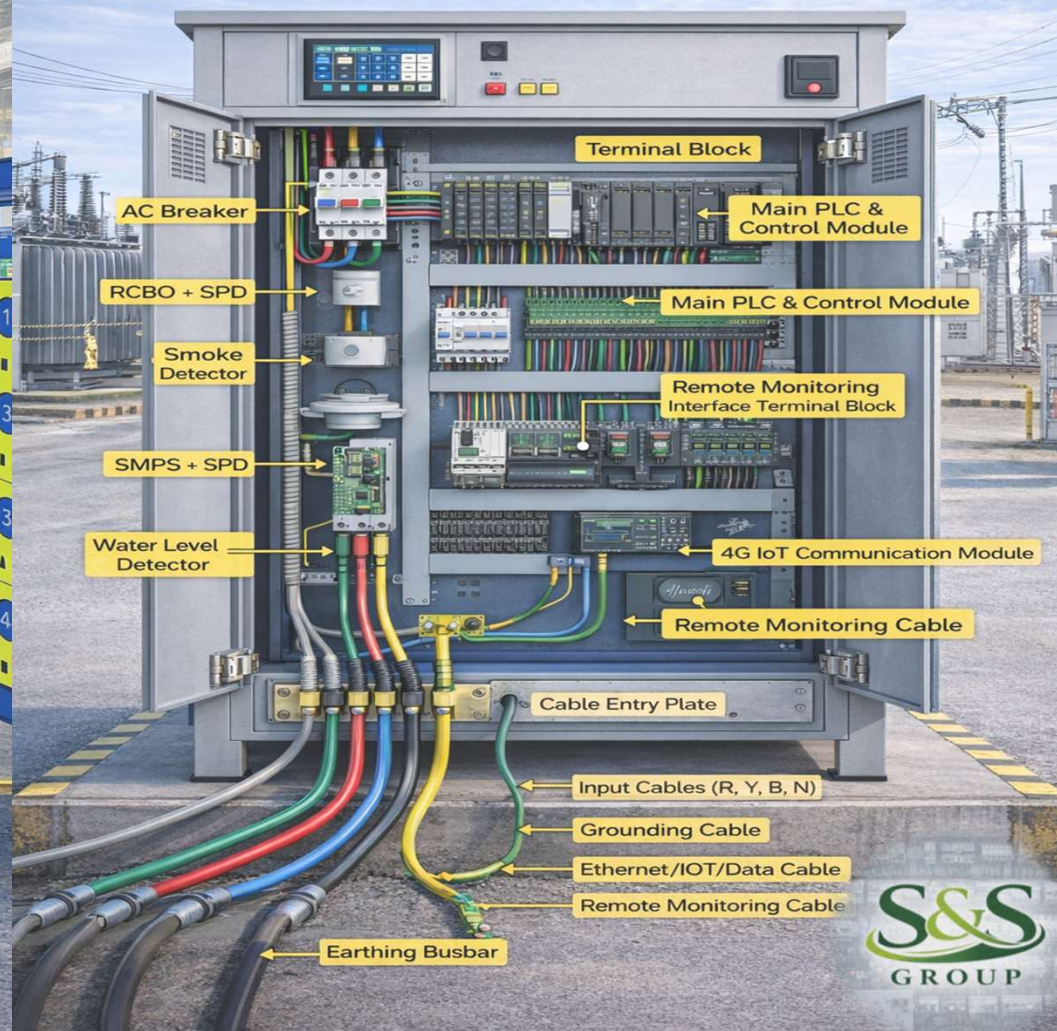


Power Supply Cable Entry

R	Red	V/G	Blue
Y	Yellow	Y/G	Yellow/Green
B	Blue	BATT+	BATT+ ('BAT1-')

Note: Designed Based On Indian Standards.

Automation & Control System & Internal Wiring & Cable Entry View



Automated Battery Swap Station

Indian Standard Based Features

Station Configuration

-  **Total Slots : 12 Nos.** (3 Rows × 4 Columns)
-  **Individual Lockable Battery Compartments**
-  **Automatic Locking System**
-  **Numbered Slots for Easy Identification**
-  **Ventilated Compartments**
-  **Powder-Coated Steel Body**
-  **Anti-Corrosion Enclosure**
-  **LED Status Indicators** (Charging / Available / Fault)



Automatic
Locking



Smoke
Detection



Water Level
Detection



Electrical
Protection

Electrical Specifications

- **Input Voltage : 150V – 270V AC**
- **Output : 230V DC / 15A per Slot**
- **Frequency : 50 Hz**
- **Power Factor : > 0.92**
- **Weight : Approx. 360 kg**

Connectivity Options

Advanced Communication for Remote Monitoring



4G



IoT



RFID



NFC



Cloud Monitoring



Remote
Monitoring



4G IoT
Communication



RFID / NFC
Access



Data Logging



ELECTRICAL SPECIFICATIONS

ELECTRICAL SPECIFICATIONS

Parameter	Specification
✓ Input Voltage	150V – 270V AC
✓ Current Rating	115 Amp
✓ Power Factor	≥ 0.92
✓ Frequency	50 Hz
✓ Charging Type	Smart Battery Charging System

Stable power input with efficient energy usage and intelligent charging management.

Heavy-duty industrial grade sheet metal cabinet for indoor and semi-outdoor installation (manufactured as per Indian Standards).

Heavy-duty industrial grade sheet metal cabinet for indoor and semi-outdoor installation (manufactured as per Indian Standards).

PHYSICAL DIMENSIONS



► Length	1866 mm
► Width	744 mm
► Height	1600 mm
► Weight	Approx. 360 kg

DISPLAY & MONITORING SYSTEM

10.1 Inch Touch Display Panel Features



Additional LED Indicators

- Charging Status
- Battery Availability
- Fault Indication
- User Authentication



OPERATING CONDITIONS

- | Parameter | Range |
|----------------------------------|-------------|
| • Operating Temperature | 0°C to 45°C |
| • Protection Cut-off Temperature | Above 50°C |



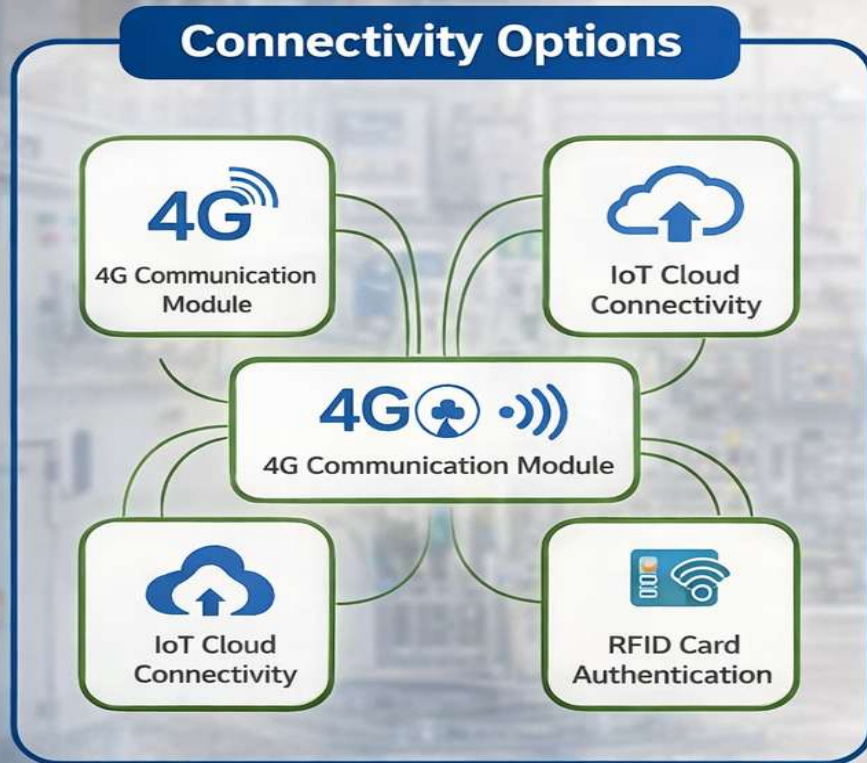
- The station is designed for reliable operation in various environmental conditions.

Parameter

- Operating Temperature
0°C to 45°C
- Protection Cut-off Temperature



Connectivity Options



Safety Mechanisms



*Indian Standards Based Safety & Healthcare Requirements

EARTHING, TESTING & INSPECTION – Automated Battery Swap Station



Indian Standards Based (IS)

EARTHING SYSTEM (IS 3043 / IS 732)



INSULATION RESISTANCE TEST (Megger – IS 732 / IS 1680)



CONTINUITY TEST (Earth Conductor – IS 732)



RCD / MCB TESTING – IS/IEC 60947



VISUAL INSPECTION & LABELING



TESTING LOG & DOCUMENTATION



Standards Followed: IS 3043 IS 732 IS 733 IS 1680 IS/IEC 60841 IS/IEC 61851

Structure & Cabinet Construction & Applications

Structure & Cabinet Construction

• 10.1 Inch Touch Display Panel

• Modular Battery Compartments

• LED Indicator

• Powder-coated Steel Enclosure

Applications

• Electric Two-Wheeler Charging Infrastructure

• Electric Three-Wheeler Fleet Stations

• Commercial EV Charging Networks

• Electric Two-Wheeler Fleet Stations

• Urban Mobility Hubs

Structure & Cabinet Construction

- ✓ Powder-coated steel enclosure
- ✓ Modular battery compartments
- ✓ Front Access Doors for battery insertion
- ✓ Integrated control panel Section
- ✓ Anti-corrosion surface finish
- ✓ Front Access Doors
- ✓ Integrated Control Panel Section
- ✓ Ventilation Arrangement
- ✓ Anti-corrosion surface finish

Structure & Cabinet Construction & Applications



DOCUMENTATION

- ✓ User Manual
- ✓ Installation Manual
- ✓ Test Report
- ✓ Commissioning Report
- ✓ Maintenance Guidelines



SCOPE INCLUDES

- ✓ Automated Battery Swap Station Enclosure
- ✓ Battery Charging Racks
- ✓ Smart EV Chargers
- ✓ Control System (PLC + HMI)
- ✓ Electrical Protection Devices
- ✓ Busbars & Internal Wiring



SCOPE INCLUDES

- ✓ Automated Battery Swap Station Enclosure
- ✓ Battery Charging Racks
- ✓ Smart EV Chargers
- ✓ Control System (PLC + HMI)
- ✓ Electrical Protection Devices
- ✓ Busbars & Internal Wiring
- ✓ All Installation Accessories



WARRANTY

- ✓ 12 Months Standard Warranty
- ✓ Manufacturing Defect Coverage

